$k = 2,$ $\quad z = \cos\pi + i\sin\pi = -1.$

$k = 3,$ $\quad z = \cos\dfrac{3\pi}{2} + i\sin\dfrac{3\pi}{2} = -\cos\dfrac{\pi}{2} - i\sin\dfrac{\pi}{2} = -i.$

Fourth roots of unity are $1, i, -1, -i \Rightarrow 1, \omega, \omega^2,$ and ω^3, where $\omega = e^{i\frac{2\pi}{4}} = i.$

Note

(i) In this chapter the letter ω is used for n^{th} roots of unity. Therefore the value of ω is depending on n as shown in following table.

value of n	2	3	4	5	⋯	k
value of ω	$e^{i\frac{2\pi}{2}}$	$e^{i\frac{2\pi}{3}}$	$e^{i\frac{2\pi}{4}}$	$e^{i\frac{2\pi}{5}}$	⋯	$e^{i\frac{2\pi}{k}}$

(ii) The complex number $z\,e^{i\theta}$ is a rotation of z by θ radians in the counter clockwise direction about the origin.

Example 2.34

Solve the equation $z^3 + 8i = 0$, where $z \in \mathbb{C}$.

Solution

Let $z^3 + 8i = 0$. Then, we get

$$z^3 = -8i$$

$$= 8(-i) = 8\left(\cos\left(-\frac{\pi}{2} + 2k\pi\right) + i\sin\left(-\frac{\pi}{2} + 2k\pi\right)\right), k \in \mathbb{Z}.$$

Therefore, $z = \sqrt[3]{8}\left(\cos\left(\dfrac{-\pi + 4k\pi}{6}\right) + i\sin\left(\dfrac{-\pi + 4k\pi}{6}\right)\right),\ k = 0,1,2.$

Taking $k = 0,1,2$, we get,

$k = 0,$ $\quad z = 2\left(\cos\left(-\dfrac{\pi}{6}\right) + i\sin\left(-\dfrac{\pi}{6}\right)\right) = 2\left(\dfrac{\sqrt{3}}{2} - i\dfrac{1}{2}\right) = \sqrt{3} - i$

$k = 1,$ $\quad z = 2\left(\cos\left(\dfrac{\pi}{2}\right) + i\sin\left(\dfrac{\pi}{2}\right)\right) = 2 = 2(0 + i) = 0 + 2i = 2i.$

$k = 2,$ $\quad z = 2\left(\cos\left(\dfrac{7\pi}{6}\right) + i\sin\left(\dfrac{7\pi}{6}\right)\right) = 2\left(\cos\left(\pi + \dfrac{\pi}{6}\right) + i\sin\left(\pi + \dfrac{\pi}{6}\right)\right)$

$$= 2\left(-\cos\left(\frac{\pi}{6}\right) - i\sin\left(\frac{\pi}{6}\right)\right) = 2\left(-\frac{\sqrt{3}}{2} - i\frac{1}{2}\right) = -\sqrt{3} - i.$$

The values of z are $\sqrt{3} - i,\ 2i,$ and $-\sqrt{3} - i.$

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